Agenda

1. Announcements
   - all early assignments accepted through February 5
   - discussion quiz 1 on Thursday
   - Check for exam conflicts and fill out the form
   - Math Learning Center opens today!
2. Wrap up Integration by Parts
3. Trig Integrals

Find a definite integral using Integration by Parts

~~Method 1~~: Find the indefinite integral and then use the Fundamental Theorem of Calculus

Method 2: Find the definite integral directly by using the limits of integration at each step of the process. That is,

I will always use Method 2

$$\boxed{\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du}$$

Example: $\int_0^{\pi/4} x\sec^2 x\,dx$

$u = x \quad du \longrightarrow dx$

$dv = \sec^2 x\,dx \longrightarrow \tan x$

$$= x\tan x\Big|_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \tan x\,dx$$

$$= \left(\frac{\pi}{4}\cdot\tan\left(\frac{\pi}{4}\right) - 0\right) - \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x}\,dx \qquad \begin{array}{l} w = \cos x \\ dw = -\sin x \end{array}$$

$$= \frac{\pi}{4} + \int_{\cos(0)\ 1}^{\cos(\frac{\pi}{4})\ \frac{\sqrt{2}}{2}} \frac{dw}{w} = \frac{\pi}{4} + \Big[\ln|w|\Big]_1^{\frac{\sqrt{2}}{2}}$$

$$= \frac{\pi}{4} + \left(\ln\frac{\sqrt{2}}{2} - \cancel{\ln 1}\right)$$

$$= \boxed{\frac{\pi}{4} + \ln\frac{\sqrt{2}}{2}}$$

~~Reduction Formulas~~

not on exams

Example: Prove the reduction formula

$$\int x^n \arcsin x \, dx = \frac{x^{n+1}}{n+1} \arcsin x - \frac{1}{n+1} \int \frac{x^{n+1}}{\sqrt{1-x^2}} \, dx$$

$u = \arcsin x \qquad dv = x^n dx$

$du = \frac{1}{\sqrt{1-x^2}} dx \qquad v = \frac{1}{n+1} x^{n+1}$

$$\int x^n \arcsin x \, dx = \frac{x^{n+1}}{n+1} \arcsin x - \int \left(\frac{x^{n+1}}{n+1}\right)\left(\frac{1}{\sqrt{1-x^2}} dx\right)$$

$$\int x^n \arcsin x \, dx = \frac{x^{n+1}}{n+1} \arcsin x - \frac{1}{n+1} \int \frac{x^{n+1}}{\sqrt{1-x^2}} \, dx$$

Trigonometric Integrals

Trigonometric Formulas

$$\sin(2x) = 2\sin(x)\cos(x)$$

$$\cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\sin^2(x) = \frac{1-\cos(2x)}{2}$$

$$\cos^2(x) = \frac{1+\cos(2x)}{2}$$

$$\sec^2(x) - 1 = \tan^2(x)$$

Also, memorize:

$$\boxed{\sin^2(x) + \cos^2(x) = \underline{1}}$$

$$\sin(x)\cos(y) = \tfrac{1}{2}\left(\sin(x-y) + \sin(x+y)\right)$$

$$\sin(x)\sin(y) = \tfrac{1}{2}\left(\cos(x-y) - \cos(x+y)\right)$$

$$\cos(x)\cos(y) = \tfrac{1}{2}\left(\cos(x-y) + \cos(x+y)\right)$$

Integrals with Powers of Sine and/or Cosine

Examples

1. $\int \cos^2 t\,dt = \int \frac{1+\cos(2t)}{2} = \frac{1}{2}\int 1+\cos 2t\,dt$

$= \boxed{\frac{1}{2}\left[t+\frac{1}{2}\sin 2t\right]+C}$

★ 2. $\int \sin^5\theta\,d\theta = \int \sin^4\theta\,\sin\theta$

goal: make $-du = \sin\theta$

$\rightarrow (u = \cos\theta)$

$= \int (\sin^2\theta)^2 \sin\theta\,d\theta$

$= \int (1-\cos^2\theta)^2 \sin\theta\,d\theta$

$= \int (1-u^2)^2\,du$

$= -\int (1-2u^2+u^4)\,du$

$= \left[-u+\frac{2}{3}u^3-\frac{u^5}{5}\right]+C$

$\rightarrow \boxed{-\cos\theta+\frac{2}{3}\cos^3\theta-\frac{1}{5}\cos^5\theta+C}$

Summary: Integrals of the form $\int \sin^n x\,dx$ OR $\int \cos^n x\,dx$, $n \geq 2$

	Integral	Use the Identity	u-substitution
n odd	$\int \sin^n x\,dx = \int \sin^{n-1} x \sin x\,dx$	$\sin^2 x = 1-\cos^2 x$	$u = \cos x;\ du = -\sin x\,dx$
	$\int \cos^n x\,dx = \int \cos^{n-1} x \cos x\,dx$	$\cos^2 x = 1-\sin^2 x$	$u = \sin x;\ du = \cos x\,dx$
n even	$\int \sin^n x\,dx = \int \left[\frac{1-\cos(2x)}{2}\right]^{n/2} dx$	$\sin^2 x = \frac{1-\cos(2x)}{2}$	$u = 2x;\ du = 2\,dx$
	$\int \cos^n x\,dx = \int \left[\frac{1+\cos(2x)}{2}\right]^{n/2} dx$	$\cos^2 x = \frac{1+\cos(2x)}{2}$	$u = 2x;\ du = 2\,dx$

Activity: $\int \sin^3 x \sqrt{\cos x}\, dx$

$$= \int \sin^2 x \cos^{1/2} x \sin x \, dx$$

make u cosx

$$-\sin x\, dx = du$$
$$-du = \sin x\, dx$$

$$= \int (1 - \cos^2 x) \cos^{1/2} x \sin x$$

$$= -\int (1 - u^2) u^{1/2}\, du$$

$$= -\int u^{1/2} - u^{3/2}\, du \quad -\left[\frac{2}{3}u^{3/2} - \frac{2}{7}u^{7/2} + C\right]$$

$$\boxed{-\frac{2}{3}\cos x^{3/2} + \frac{2}{7}\cos^{7/2} x + C}$$

Example: $\int \sin^2 x \cos^4 x \, dx$

$$\int (1 - \cos^2 x) \cos^4 x \, dx$$

$$\int \cos^4 x - \cos^6 x \, dx$$

$$\int \cos^4 x\, dx - \int \cos^6 x\, dx$$

$\vdots$

(use powers of cosine table)

Summary of integrals of the form $\int \sin^m x \cos^n x\, dx$

	Integral	Use the Identity	u-substitution
m odd	$\int \sin^m x \cos^n x\, dx = \int \sin^{m-1} x \cos^n x \sin x\, dx$	$\sin^2 x = 1 - \cos^2 x$	$u = \cos x$ $du = -\sin x\, dx$
n odd	$\int \sin^m x \cos^n x\, dx = \int \sin^m x \cos^{n-1} x \cos x\, dx$	$\cos^2 x = 1 - \sin^2 x$	$u = \sin x$ $du = \cos x\, dx$
m and n both even	$\int \sin^m x \cos^n x\, dx = \int (1 - \cos^2 x)^{m/2} \cos^n x\, dx$ or $\int \sin^m x \cos^n x\, dx = \int \sin^m x (1 - \sin^2 x)^{n/2}\, dx$	$\cos^2 x = \dfrac{1 + \cos(2x)}{2}$ $\sin^2 x = \dfrac{1 - \cos(2x)}{2}$	$u = 2x;\ du = 2\, dx$ $u = 2x;\ du = 2\, dx$

Integrals Involving Secant and Tangent

	The Integral	Use the Identity	Method
m odd	$\int \tan^m x \sec^n x\, dx = \int \tan^{m-1} x \sec^{n-1} x \sec x \tan x\, dx$	$\tan^2 x = \sec^2 x - 1$	$u = \sec x$ $du = \sec x \tan x\, dx$
n even	$\int \tan^m x \sec^n x\, dx = \int \tan^m x \sec^{n-2} x \sec^2 x\, dx$	$\sec^2 x = 1 + \tan^2 x$	$u = \tan x$ $du = \sec^2 x\, dx$
m even; n odd	$\int \tan^m x \sec^n x\, dx = \int \left(\sec^2 x - 1\right)^{m/2} \sec^n x\, dx$	$\tan^2 x = \sec^2 x - 1$	Use integration by parts

Example / Activities:

1.) $\int \tan^3 x \sec^4 x\, dx$

2.) $\int \tan^2 x \sec^6 x\, dx$

3.) $\int \tan^2 x \sec x \, dx$

Using Product Identities

$$\sin(x)\cos(y) = \frac{1}{2}\left(\sin(x-y) + \sin(x+y)\right)$$

$$\sin(x)\sin(y) = \frac{1}{2}\left(\cos(x-y) - \cos(x+y)\right)$$

$$\cos(x)\cos(y) = \frac{1}{2}\left(\cos(x-y) + \cos(x+y)\right)$$

Example: $\int \sin(3x)\cos(2x)\,dx$